Bibliography – FX Modelling References

This document provides a consolidated bibliography of key academic and practitioner references commonly used for foreign exchange (FX) modelling, calibration, and risk management. The references below are widely cited in front-office pricing, XVA / PFE simulation frameworks, and independent model validation.

Core FX Options and Smile Modelling

Wystup, U. (2017). *FX Options and Structured Products*. 2nd Edition, John Wiley & Sons.

A practitioner-focused reference covering FX market conventions, vanilla and exotic FX options, volatility smiles, calibration techniques, and structured product payoffs. Widely used on FX derivatives desks.

Castagna, A. (2010). *FX Options and Smile Risk*. John Wiley & Sons, Wiley Finance Series.

A detailed treatment of FX volatility smiles, smile dynamics, and hedging of smile risk. Particularly relevant for understanding model risk, hedging errors, and limitations of lognormal and local-volatility frameworks.

Gatheral, J. (2006). *The Volatility Surface: A Practitioner’s Guide*. John Wiley & Sons.

A foundational reference on implied volatility surfaces, static arbitrage constraints, and parameterisation of smiles. While not FX-specific, it is directly applicable to FX volatility modelling and validation.

Local Volatility, Stochastic Volatility, and Numerical Methods

Dupire, B. (1994). “Pricing with a Smile.” *Risk Magazine*, January 1994.

The seminal paper introducing local volatility models and the relationship between implied volatility surfaces and diffusion dynamics.

Bergomi, L. (2016). *Stochastic Volatility Modeling*. Chapman & Hall / CRC Financial Mathematics Series.

A modern reference on stochastic volatility models, volatility dynamics, and the limitations of local-volatility approaches. Particularly useful for understanding smile dynamics and tail risk.

Wilmott, P. (2006). *Paul Wilmott on Quantitative Finance*. 2nd Edition, John Wiley & Sons.

A broad reference covering stochastic calculus, PDE methods, Monte Carlo simulation, and numerical techniques widely used in derivatives modelling, including FX.

Stochastic Calculus, Measure Theory, and Arbitrage

Shreve, S. E. (2004). *Stochastic Calculus for Finance II: Continuous-Time Models*. Springer Finance.

A rigorous introduction to stochastic calculus, change of measure, numeraire techniques, and arbitrage-free pricing. Essential for understanding FX modelling under different measures.

Björk, T. (2009). *Arbitrage Theory in Continuous Time*. 3rd Edition, Oxford University Press.

A mathematically rigorous treatment of arbitrage pricing theory, martingale measures, and multi-asset modelling, including foreign exchange.

Interest Rate Modelling and FX–Rates Interaction

Brigo, D., & Mercurio, F. (2006). *Interest Rate Models – Theory and Practice*. 2nd Edition, Springer Finance.

A comprehensive reference on interest rate modelling, curve construction, and multi-curve frameworks. Particularly relevant for FX forward modelling and cross-currency products.

Andersen, L. B. G., & Piterbarg, V. V. (2010). *Interest Rate Modeling*, Volumes I–III. Atlantic Financial Press.

An advanced reference on interest rate modelling, numerical methods, collateralisation, and measure consistency. Frequently used in XVA, PFE, and multi-asset simulation contexts involving FX.

Hedging and Risk Management Perspective

Taleb, N. N. (1997). *Dynamic Hedging: Managing Vanilla and Exotic Options*. John Wiley & Sons.

A practitioner-oriented book focusing on hedging nonlinear risks, model limitations, and the fragility of dynamic hedging strategies. Useful for qualitative model risk discussions.

Usage in Model Validation and PFE Context

These references collectively support: - conceptual justification of FX diffusion models (GBM, local/stochastic volatility); - calibration methodologies for FX volatility and forward-implied drift; - assessment of model limitations (smile risk, tail behaviour, regime shifts); - validation of measure consistency and arbitrage-free dynamics; - governance, monitoring, and model risk management in PFE and XVA frameworks.

This bibliography is suitable for inclusion in independent model validation reports, regulatory submissions, and internal model documentation.